Photonic-crystal ring resonators with controlled mode splitting, contra-directional couplers and MZIs

edX Phot1x / openEBL-2026-10: design report for layout `EBeam_zeyad_874`

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Abstract

This chip (586 µm × 401 µm, 12 test structures, 36 grating couplers) studies three things. (1) Six photonic-crystal ring resonators (PhCRs, $R = 10$ µm) with N square notches on the sidewall. The notch count N selects which resonance splits ($m = N/2$), and the notch depth sets the size of the split. Two families are made: 30 nm notches on the inner wall ("in30") and 15 nm notches on both walls ("dual15"), each at $N = 196, 198, 200$. Bloch-cell and 3D FDTD predict ring splits of 4.2 nm to 5.5 nm, more than 25 linewidths and below the free spectral range (FSR ≈ 9.2 nm). (2) Two contra-directional couplers (contra-DCs) with grating periods 316 and 320 nm, which move the drop band from 1544 nm to 1555.5 nm. (3) Three MZIs with $\Delta L = 100, 150, 200$ µm, used to extract the group index; simulation gives $n_g = 4.19 \pm 0.02$. The layout passed the openEBL submission checks and functional/DFT verification with 0 errors and was merged (PR #970).

1 Introduction and goals

A ring resonator supports two degenerate travelling-wave modes at each resonance: clockwise (CW) and counter-clockwise (CCW). A small periodic perturbation of the sidewall couples them, and the resonance splits into two standing-wave modes [1]. Random roughness does this weakly and at random resonances. Here the perturbation is designed: N identical notches around the ring. The goals of the chip are:

1. **Position control:** show that N chooses which azimuthal order m splits, so the split moves by one FSR per $\Delta N = 2$.
2. **Strength control:** compare one deep notch (30 nm, inner wall) with two shallower notches (15 nm + 15 nm, both walls) that remove the same silicon.
3. **Fabrication feedback:** use the measured split to estimate the depth that was actually printed.
4. **Reference devices:** contra-DC period study [2], MZIs for n_g, and a loopback for the grating-coupler envelope.

2 Theory

2.1 MZI

The phase difference between the arms is $\Delta\phi = 2\pi n_{\text{eff}}(\lambda)\Delta L/\lambda$. Neighbouring transmission nulls differ by 2π. Differentiating, including the dispersion of n_{eff}, gives

$$\text{FSR} = \frac{\lambda^2}{n_g\,\Delta L}, \qquad n_g = n_{\text{eff}} - \lambda\frac{dn_{\text{eff}}}{d\lambda}. \tag{1}$$

All MZIs use the same waveguide, so $\mathrm{FSR} \cdot \Delta L = \lambda^2/n_g$ must be the same for all three. This is the check used in Sec. 4.

2.2 Mode splitting in a notched ring

A ring of radius R has resonances at $m\lambda_m = 2\pi R\, n_{\mathrm{eff}}$. The N notches form a grating with period $\Lambda = 2\pi R/N$ along the centre line. CW and CCW light couple when the Bragg condition $2\beta = 2\pi/\Lambda$ holds, which gives

$$m = N/2. \tag{2}$$

Therefore $N = 200, 198, 196$ split the orders $m = 100, 99, 98$, and each $\Delta N = 2$ moves the split by one FSR to longer wavelength. The split size is proportional to the coupling coefficient, i.e. to the *first Fourier harmonic* of the wall profile. For a 50% duty square notch of depth D this harmonic is $(4/\pi)(D/2) = 2D/\pi$, while a sine of the same depth gives $D/2$. So if lithography rounds the square notches fully into a sine, the split drops at most to $\pi/4 \approx 0.79$ of the design value.

2.3 Contra-directional coupler

Forward light in waveguide 1 couples to backward light in waveguide 2 when $\beta_1 + \beta_2 = 2\pi/\Lambda$ [2]:

$$\lambda_D = \Lambda\,(n_1 + n_2), \qquad \frac{d\lambda_D}{d\Lambda} = \frac{(n_1 + n_2)^2}{n_{g1} + n_{g2}}. \tag{3}$$

With $n_1 + n_2 = \lambda_D/\Lambda \approx 4.89$ and $n_{g1} + n_{g2} \approx 8.4$, the slope is about 2.85 nm of band shift per nm of period. The two waveguides have different widths (560/440 nm), which moves their self-reflections ($\lambda = 2n_i\Lambda$) away from the drop band.

3 Chip layout

All grating couplers are `ebeam_gc_te1550`, facing right, on a 127 µm vertical pitch (rows at $y =$ 14.36, 141.36, 268.36, 395.36 µm). Columns are at least 60 µm apart. Every `opt_in` label sits on the input coupler (fibre 2), and outputs use fibres 1, 3 and 4. Labels follow the pattern `opt_in_TE_1550_device_zeyad874_<device>`.

Table 1: Devices, label names and coupler positions (input coupler x, y in µm).

Device label	Type	Input GC	Outputs (fibre)	Key parameter
`MZI100`	MZI	(40.9, 395.36)	3	$\Delta L = 100$ µm
`MZI150`	MZI	(40.9, 141.36)	3	$\Delta L = 150$ µm
`MZI200`	MZI	(120.9, 395.36)	3	$\Delta L = 200$ µm
`loopback`	GC to GC	(120.9, 141.36)	3	reference
`PhCRin30N196/198/200`	PhC ring	(196/256/316, 268.36)	1 drop, 3 through, 4 add	inner wall, 30 nm
`PhCRdual15N196/198/200`	PhC ring	(376/436/496, 268.36)	1 drop, 3 through, 4 add	both walls, 15 nm
`CDC5025p316`	contra-DC	(557, 141.36)	3 drop	$\Lambda = 316$ nm
`CDC5025p320`	contra-DC	(557, 395.36)	3 drop	$\Lambda = 320$ nm

4 MZIs and loopback

The MZIs use two `ebeam_y_1550` splitters and 500 nm × 220 nm strip waveguides (TE). The arm lengths are 159.397 (short) and 259.397 / 309.397 / 359.397 µm (long). Each circuit was simulated in Lumerical INTERCONNECT, and the spectra were analysed with a script that locates the transmission nulls. The script computes the FSR and n_g at 1550 nm (Eq. 1) and the extinction ratio (ER).

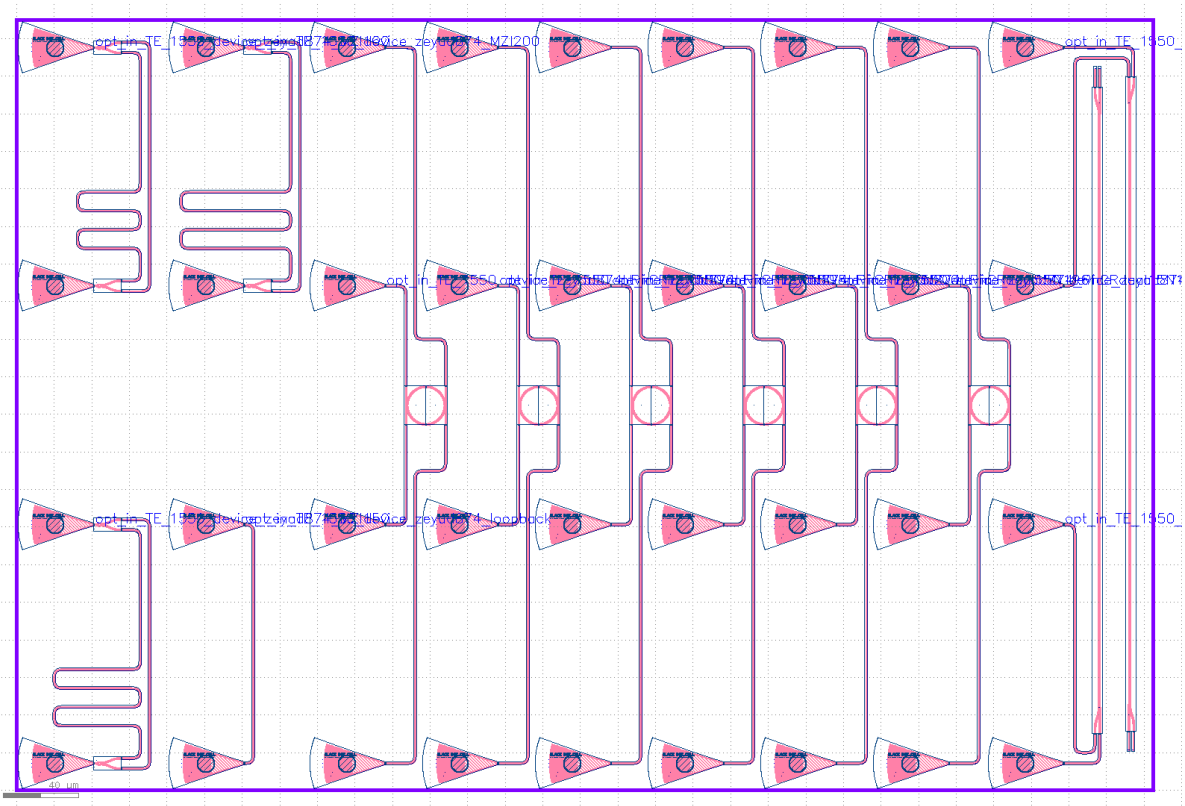


Figure 1: Full layout of `EBeam_zeyad_874` (KLayout). Left: MZIs and loopback. Centre: six PhC rings. Right: two contra-DCs.

Table 2: MZI results at 1550 nm (simulation).

	FSR (nm)	n_g	FSR·ΔL (nm µm)	$\mathrm{FSR}_{100}/\mathrm{FSR}$	ER (dB)	Peak (dB)
MZI100	5.745	4.182	574.5	1.000	37.3	−5.15
MZI150	3.837	4.174	575.6	1.497	34.4	−5.19
MZI200	2.853	4.211	570.6	2.014	41.1	−5.18

The ratios match the expected 1.5 and 2 within 0.7%, and $n_g = 4.19 \pm 0.02$ is the same for all three, so the arm lengths are correct. The remaining spread comes from locating the nulls on the 50 pm simulation grid, and it is largest for the smallest FSR (MZI200). The extinction ratio of 34 to 41 dB shows near-ideal 50/50 splitting. The peak of about −5.2 dB is almost entirely grating-coupler loss. The loopback peaks at about −4.9 dB near 1552 to 1555 nm, about 2.4 dB per coupler, with a 3 dB band of roughly 1530 to 1576 nm (Fig. 3). All measured spectra will be normalised by the loopback.

5 Photonic-crystal rings

5.1 Design

Each ring has $R = 10$ µm, width 500 nm, and a 125 nm gap to two vertical bus waveguides (add-drop). It is built from two identical half-ring cells, the same geometry that was simulated in 3D FDTD. The notches are square with 50% duty and about 157 nm long. A full (un-notched) section is centred on each coupling point. The port map is: input GC268, through GC141, drop GC395, add GC14.

A ray-trace check of the final layout confirmed the following for every ring: N notches at the designed depth (30 nm on the inner wall, or 15 nm on both walls), $R_{\mathrm{out}} = 10.251$ µm, width 500 nm,

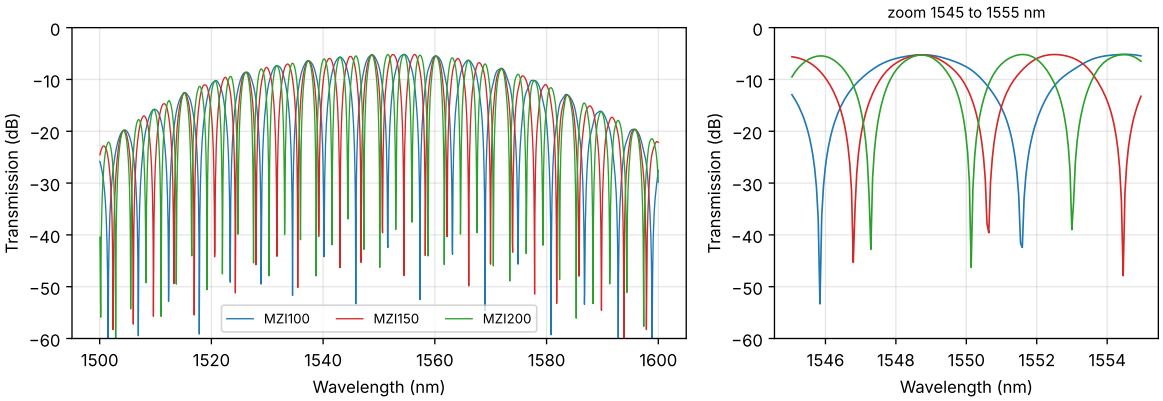


Figure 2: Simulated MZI transmission (INTERCONNECT, including both grating couplers). The FSR scales as $1/\Delta L$.

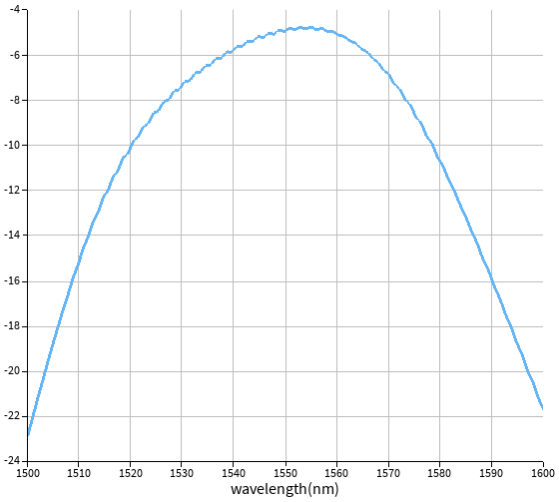


Figure 3: Loopback (two grating couplers), INTERCONNECT.

and a gap of 124 to 125 nm (1 nm layout grid). The two half rings form one closed polygon.

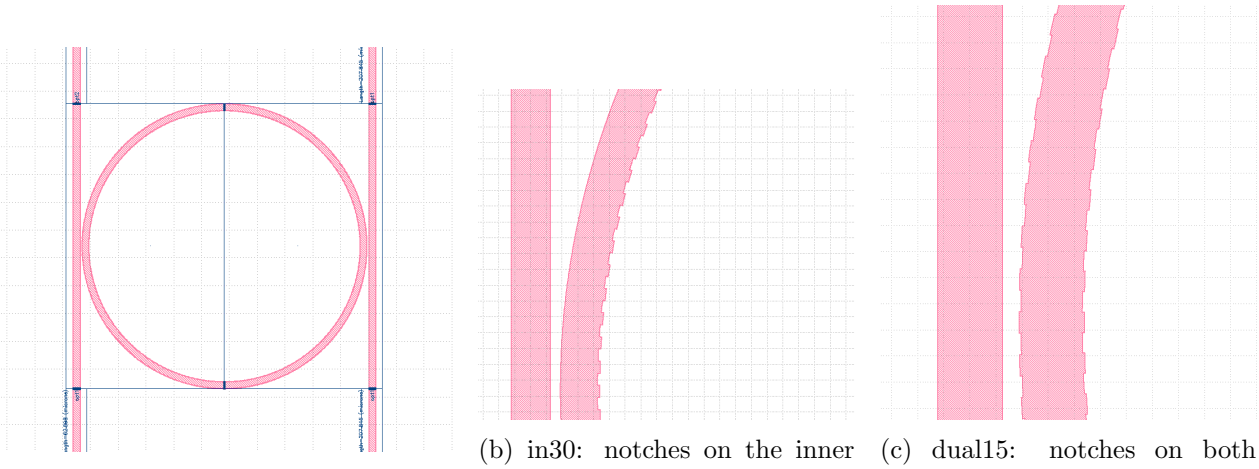
5.2 Simulation method

- **Bloch-cell FDTD (Tidy3D):** one period of a straight notched waveguide with a Bloch boundary at $k = \pi/\Lambda$, i.e. the band edge. This is the same method as the SiEPIC Bragg-grating workshop [3]. The two band-edge modes give the split and its centre. Mesh: 2 nm (in30) or 1 nm (dual15) across the walls; all edges are on grid lines.
- **3D half-ring FDTD:** the real curved half ring with its bus. This gives a 4-port S-matrix, which is joined into a full ring and exported to INTERCONNECT.
- **Bend factor:** ring split divided by straight split. Measured as 3D/Bloch: 0.936 for in30 at $N = 200$ (used in Table 3), 0.94 from the earlier inner-wall 10 nm runs, and 0.964 for notches on both walls (measured on dual10).

5.3 Results

Main observations:

- **Position control:** going from $N = 198$ to 200 moves the split by 9.22 nm (one FSR) for both families, as expected from $m = N/2$.



(a) One ring with its two buses.

(b) in30: notches on the inner wall.

(c) dual15: notches on both walls.

Figure 4: PhC ring layout (KLayout).

Table 3: Predicted split and centre wavelength. Straight values: Bloch FDTD. Ring values: straight × bend factor. †extrapolated (one FSR per $\Delta N = 2$), Bloch run pending.

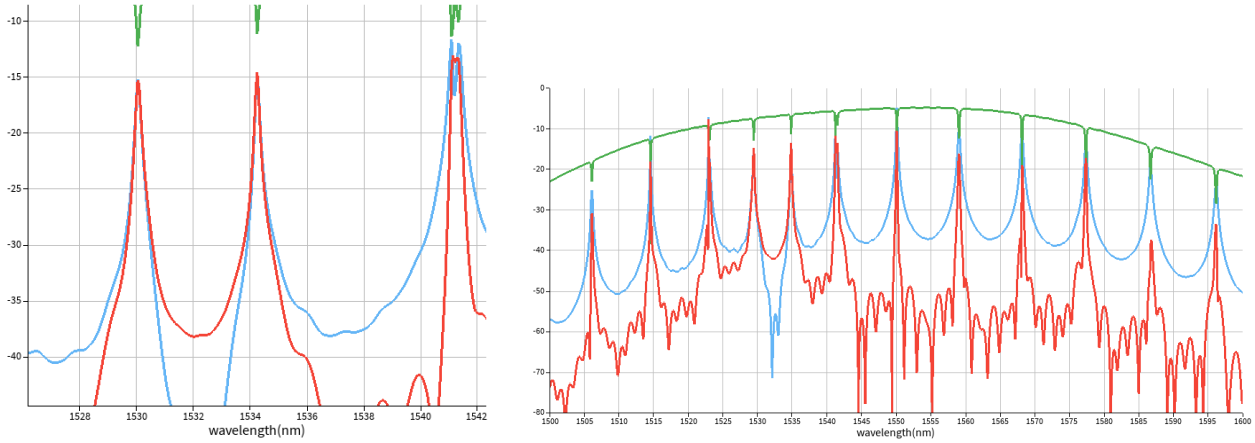
Ring	m	Centre (nm)	Split, straight (nm)	Split, ring (nm)
in30 N196	98	$\approx 1549.5^{\dagger}$	$\approx 4.65^{\dagger}$	$\approx 4.35^{\dagger}$
in30 N198	99	1540.27	4.566	4.27
in30 N200	100	1531.05	4.486	4.20
dual15 N196	98	$\approx 1550.2^{\dagger}$	$\approx 5.73^{\dagger}$	$\approx 5.5^{\dagger}$
dual15 N198	99	1541.02	5.639	5.44
dual15 N200	100	1531.80	5.551	5.35

- **Independent check:** for in30 N200, the 3D half-ring gives a 4.199 nm split at 1532.1 nm, against 4.216 nm (Bloch × 0.94, the bend factor from the earlier 10 nm runs) and 1531.05 nm from Bloch. The two agree within 0.4% in split and 1 nm in centre. For dual15 N200, the 3D half-ring gives 5.399 nm at 1532.10 nm, against 5.35 nm (Bloch × 0.964) and 1531.80 nm: 0.9% and 0.3 nm.
- **Resolvability:** the split is about 28 (in30) and 35 (dual15) linewidths, so it is easy to resolve. It is also below the FSR (split/FSR ≈ 0.46 and 0.58), and the doublet stays 6.5 nm to 7 nm from the neighbouring resonances.
- **Fingerprint of the doublet:** in the 3D simulation the split resonances also reflect strongly back to the input (≈−8 dB, against −40 dB elsewhere). This identifies the CW/CCW coupled pair.
- **Two shallow notches beat one deep notch:** dual15 gives a 24% larger split than in30, with the same total silicon removed. The split grows sub-linearly with depth (it saturates), so two 15 nm steps couple more strongly than one 30 nm step.

6 Contra-directional couplers

Both devices use the PDK `contra_directional_coupler`: $w_1/w_2 = 560/440$ nm, gap 100 nm, $N = 1000$ periods, anti-reflection on, Gaussian apodisation (index 10), and corrugations 50/25 nm. The periods are 316 and 320 nm. The drop port is measured, and the through and add ports are terminated.

The band moves 11.6 nm for a 4 nm period change, about 2.9 nm/nm, in agreement with Eq. (3). After subtracting the loopback, the drop efficiency is close to 0 dB. Note: the PDK model data for



(a) in30 N200, zoom: doublet at 1530.0 / 1534.2 nm.

(b) dual15 N200, 1500 to 1600 nm: the split resonance near 1532 nm.

Figure 5: INTERCONNECT circuit simulation of the full add-drop rings: two copies of the 3D-FDTD half-ring S-matrix, plus grating couplers. Traces: through, drop and add ports.

Table 4: Contra-DC drop band.

Period	PDK data (nm)	INTERCONNECT (nm)	Flat-top width (nm)
316 nm	1543.97	≈ 1543.8	≈ 4
320 nm	1555.45	≈ 1555.5	≈ 4.5

$\Lambda = 320$ nm covers only 1530 to 1580 nm.

7 Verification and test plan

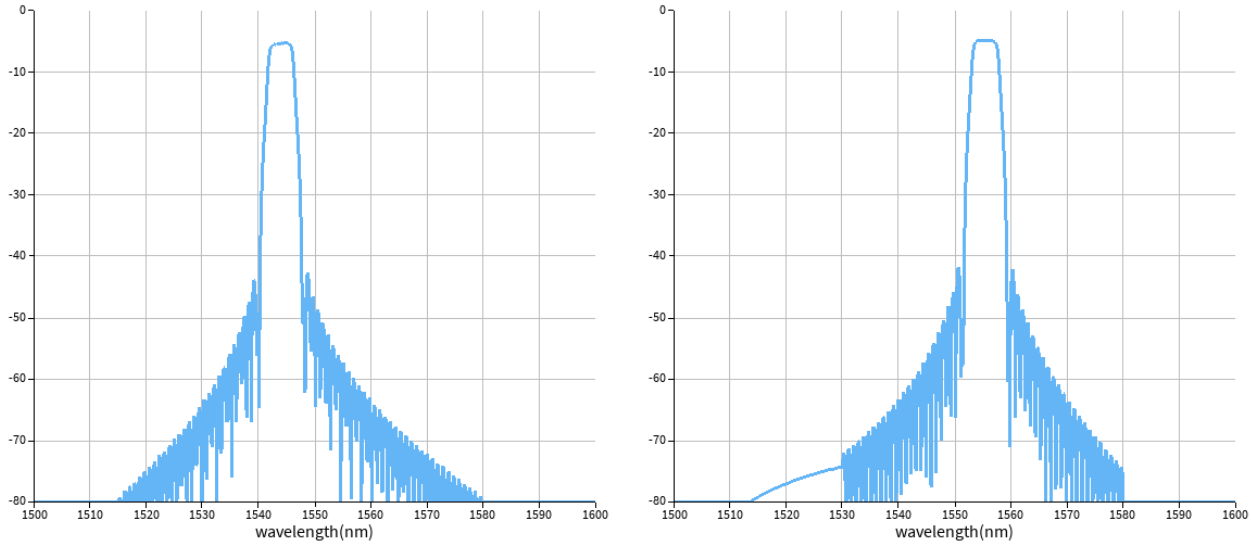
Verification. The openEBL submission checks report a bounding box of 586.153 µm × 401.228 µm (limit 605 × 410), 36 black-box couplers and 0 unreplaced cells. Functional verification, including the DFT rules, gives 0 errors. All 12 labels are unique. The layout was merged into the run in PR #970 [4].

Measurement. TE laser sweep 1499 to 1601 nm (PDK DFT rules) on the automated setup; a second set at about 10 °C higher will be requested. Each spectrum will be normalised by the loopback. Analysis plan:

1. MZIs: FSR and n_g from the nulls (Eq. 1), compared with Table 2.
2. Rings: locate the doublet and confirm $m = N/2$ from its position. Then convert the measured split to an effective printed depth with the FDTD split-vs-depth curve, and compare in30 with dual15. At the higher temperature, both doublet modes should shift together while the split stays constant.
3. Contra-DCs: band centre and width against the PDK model; centre shift versus period.

8 Conclusion

The chip combines a designed mode-splitting experiment with reference devices. Simulation predicts clearly resolved ring doublets at the order chosen by N (4.2 nm to 5.5 nm split). It also predicts a period-controlled contra-DC drop band (1544 nm to 1555.5 nm) and a consistent group index from three MZIs ($n_g = 4.19$). The measured ring splits will also measure how deep small sidewall steps print on this process.



(a) $\Lambda = 316\,\text{nm}$. (b) $\Lambda = 320\,\text{nm}$.

Figure 6: Contra-DC drop-port response (INTERCONNECT, PDK compact model, including grating couplers).

References

[1] B. E. Little, J.-P. Laine, S. T. Chu, "Surface-roughness-induced contradirectional coupling in ring and disk resonators," *Opt. Lett.* 22(1), 4-6 (1997).

[2] W. Shi, X. Wang, C. Lin, H. Yun, Y. Liu, T. Baehr-Jones, M. Hochberg, N. A. F. Jaeger, L. Chrostowski, "Silicon photonic grating-assisted, contra-directional couplers," *Opt. Express* 21(3), 3633-3650 (2013).

[3] SiEPIC, "SiEPIC Bragg workshop," `https://github.com/SiEPIC/SiEPIC_Bragg_workshop`.

[4] SiEPIC, "openEBL-2026-10," `https://github.com/SiEPIC/openEBL-2026-10`.

[5] L. Chrostowski, M. Hochberg, *Silicon Photonics Design*, Cambridge University Press (2015).